Below is a **comprehensive equipment list** for a **500 MW/year polycrystalline silicon solar cell manufacturing plant** (from incoming wafers to finished solar cells). I've also included the **testing and quality control equipment** that is typically required.

A. Wafer Receiving & Inspection

No.	Machine	Qty	Function
A1	Automatic Wafer Loader	2	Load wafer cassettes
A2	Wafer Thickness Measurement	1	Thickness measurement
A3	Wafer Resistivity Tester	1	Measure resistivity
A4	Wafer Lifetime Tester	1	Minority carrier lifetime
A5	Wafer Flatness Tester	1	Warp & bow measurement
A6	Automatic Optical Inspection (AOI)	2	Crack & defect inspection
A7	Barcode/QR Identification	2	Product traceability

B. Texturing Line

No.	Machine	Qty
B1	Automatic Cassette Loader	2
B2	Alkaline Texturing Machine	2
B3	DI Water Rinse	2
B4	Drying Machine	2
B5	Automatic Unloader	2

Chemicals:

- NaOH
- KOH
- IPA
- DI water

C. Cleaning Line

No.	Machine	Qty
C1	RCA Cleaning Line	2
C2	HF Cleaning Tank	2
C3	HCl Cleaning Tank	2
C4	DI Water Cascade	2
C5	Dryer	2

D. Phosphorus Diffusion

No.	Machine	Qty
D1	Automatic Boat Loader	2
D2	Quartz Boat Cleaner	1
D3	$POCl_3$ Diffusion Furnace	8
D4	Nitrogen Supply System	1
D5	Oxygen Supply System	1
D6	$POCl_3$ Bubbling Cabinet	2
D7	Furnace Cooling Station	2

E. PSG Removal

No.	Machine	Qty
E1	HF Etching Machine	2
E2	Cascade Washing	2
E3	Dryer	2

F. Laser Edge Isolation

No.	Machine	Qty
F1	UV/Fiber Laser Isolation	4
F2	Dust Collection System	4

G. PECVD

No.	Machine	Qty
G1	PECVD Reactor	6
G2	Vacuum Pump System	6
G3	Silane Gas Cabinet	2
G4	Ammonia Gas Cabinet	2
G5	Nitrogen Cabinet	1

H. Screen Printing

Front Silver

Machine	Qty
Automatic Screen Printer	4

Rear Aluminum

Machine	Qty
Screen Printer	4

Rear Silver

Machine	Qty
Screen Printer	4

Auxiliary:

- Paste Stirring Machine
- Screen Cleaning Machine
- Screen Inspection Microscope

I. Drying

Machine	Qty
Infrared Dryer	4

J. Firing

Machine	Qty
Belt Firing Furnace	4
Cooling Conveyor	4

K. Cell Testing

Electrical Testing

Machine	Qty	Function
Solar Cell I-V Tester	4	Efficiency, Voc, Isc, Pmax
Cell Sorting Machine	4	Automatic grading
Suns-Voc Tester	1	Advanced electrical characterization
Dark I-V Tester	1	Leakage current
Quantum Efficiency (QE) System	1	Spectral response
Sheet Resistance Tester	2	Emitter resistance
Contact Resistance Tester	2	Metal contact quality

Optical Inspection

Machine	Qty
Electroluminescence (EL) Tester	4
Infrared Camera	2
AOI Surface Inspection	2
Color Inspection	1

Mechanical Testing

Machine	Qty
Cell Strength Tester	1
Three-Point Bending Tester	1
Micro Crack Tester	2

Material Analysis Laboratory

Machine	Qty
SEM (Scanning Electron Microscope)	1
EDS Detector	1
XRD	1
Raman Spectrometer	1
FTIR	1
Optical Microscope	2

Chemical Laboratory

Machine	Qty
ICP-OES	1
UV-Vis Spectrometer	1
pH Meter	2
Conductivity Meter	2
Chemical Analyzer	1

Process Monitoring

Machine	Qty
Film Thickness Measurement	2
Ellipsometer	1
Reflectometer	2
Surface Profilometer	1

L. Automation

Machine	Qty
AGV (Automated Guided Vehicles)	6
Conveyor System	Complete
Robotic Loader	10
MES (Manufacturing Execution System)	1
SCADA	1

M. Utility Systems

Equipment
DI Water Plant
Wastewater Treatment
Air Compressor
Nitrogen Generator or Bulk Storage
Chilled Water Plant
Vacuum System
Chemical Distribution System
Exhaust Scrubber
HVAC Clean Room System
Emergency Power (UPS & Generator)

N. Packaging

Machine	Qty
Automatic Cell Stacker	2
Vacuum Packaging Machine	2
Barcode Printer	2
Labeling Machine	2
Palletizing Robot	1

Estimated Equipment Investment

Section	Approximate Cost (USD)
Process Equipment	30–45 million
Testing & QC	3–8 million
Laboratory	2–5 million
Automation	5–10 million
Utilities	8–15 million
Packaging	1–3 million
Total Equipment Investment	**49–86 million**

Notes

1. **Testing equipment is essential** in modern cell production. A 500 MW/year line will typically include **100% electrical testing (I-V), 100% electroluminescence (EL) inspection, and automated sorting**, while advanced analytical tools like SEM, XRD, and Raman are used mainly for process development, failure analysis, and quality assurance rather than testing every production cell.

2. If you are planning a new factory today, many manufacturers would configure the line for **TOPCon** rather than conventional polycrystalline cells, since much of the automation and utility infrastructure is similar, but the process equipment differs in the passivation and contact formation stages.